POCKETSMART AI

Your Smart Budget & Recommendation Assistant

PROJECT REPORT

A smart personal-finance assistant designed to track expenses, manage budgets, analyze spending patterns, and provide personalized recommendations.

Academic Project

1. Abstract

PocketSmart AI is an intelligent budget management application that helps users monitor their income and expenses and make better day-to-day financial decisions. The system records transactions, categorizes expenses, compares actual spending with user-defined budgets, and presents useful summaries through dashboards and charts. An AI recommendation component analyzes spending patterns and provides personalized suggestions such as reducing unnecessary expenses, adjusting category budgets, and improving savings habits.

2. Introduction

Managing personal money can be difficult when expenses are spread across food, transportation, education, shopping, subscriptions, and other categories. PocketSmart AI provides a simple centralized platform where users can record transactions and understand where their money is going. The application combines conventional budgeting features with AI-assisted recommendations to make financial information easier to understand and act upon.

3. Problem Statement

Many users do not maintain a clear record of daily spending and may notice overspending only after their money has already been used. Traditional expense trackers often provide records and charts but do not explain spending behavior or suggest practical next steps. PocketSmart AI addresses this gap by combining expense tracking, budget monitoring, analytics, alerts, and personalized recommendations.

4. Objectives

- Track income and expenses in one place.
- Automatically or manually categorize transactions.
- Allow users to create budgets for different categories.
- Monitor budget usage and identify overspending.
- Display monthly and category-wise spending analysis.
- Generate personalized AI-based recommendations.
- Help users set savings goals and monitor progress.

5. Proposed System

The proposed system consists of a user interface, transaction management module, budgeting engine, analytics layer, and AI recommendation engine. Users enter income and expenses through the application. Transactions are stored in a database and grouped by category. The budgeting engine compares spending against planned limits, while the analytics component identifies trends. The recommendation engine converts these insights into understandable suggestions.

6. Major Modules

6.1 User Management

Provides registration, login, profile management, and secure access to personal financial data.

6.2 Income & Expense Management

Allows users to add, edit, delete, and view income and expense transactions with date, amount, category, and description.

6.3 Expense Categorization

Transactions can be categorized into food, transport, education, shopping, bills, entertainment, health, and other categories. AI-assisted classification can be added for faster entry.

6.4 Budget Management

Users can define monthly or category-wise limits. The system calculates remaining budget and can generate alerts when spending approaches or exceeds a limit.

6.5 Analytics Dashboard

Displays total income, total expenses, savings, category-wise spending, monthly comparisons, and budget utilization using tables and charts.

6.6 AI Recommendation Assistant

Analyzes transaction patterns and generates recommendations based on spending trends, budget limits, and savings goals. Recommendations are informational and should not be treated as professional financial advice.

7. System Architecture

User → Web/Mobile Interface → Application Server → Database

■ Analytics Engine → AI Recommendation Engine → Personalized Suggestions

8. Functional Requirements

- User registration and authentication.
- Add, update, delete, and search transactions.
- Category management.
- Budget creation and monitoring.
- Dashboard and spending reports.
- Budget alerts and notifications.
- AI-generated recommendations.
- Savings-goal tracking.

9. Non-Functional Requirements

- Security: financial information must be protected with appropriate authentication and access controls.
- Usability: the interface should be simple enough for everyday users.
- Performance: dashboards and reports should load efficiently.
- Reliability: transactions should be stored consistently without accidental loss.
- Scalability: the design should support additional users and features.

10. Suggested Technology Stack

Layer	Suggested Technology
Frontend	HTML, CSS, JavaScript / React
Backend	Python Flask / FastAPI or Node.js
Database	MySQL / PostgreSQL / SQLite
AI/ML	Python, scikit-learn or an LLM API
Charts	Chart.js / Matplotlib
Authentication	JWT / secure session authentication

11. Database Design

Users Table

user_id (Primary Key), name, email, password_hash, created_at

Transactions Table

transaction_id (Primary Key), user_id (Foreign Key), type, amount, category, description, transaction_date

Budgets Table

budget_id (Primary Key), user_id (Foreign Key), category, limit_amount, month, year

Savings Goals Table

goal_id (Primary Key), user_id (Foreign Key), goal_name, target_amount, current_amount, target_date

12. AI Recommendation Logic

The recommendation component can calculate category spending, compare current spending with historical averages, detect budget thresholds, and identify repeated high-value or discretionary expenses. A simple rule-based system can be used for the first version, while a machine-learning or language-model component can be added later.

Example recommendation:

“Your food expenses are close to your monthly food budget. Consider setting a smaller weekly limit or reviewing frequent food-delivery purchases.”

13. Example User Workflow

1. User logs in.
2. User enters monthly income.
3. User creates category budgets.
4. User records daily expenses.
5. System categorizes and stores transactions.
6. Dashboard calculates budget usage and savings.
7. AI analyzes the spending pattern.
8. User receives recommendations and alerts.

14. Advantages

- Simple centralized expense tracking.
- Better awareness of spending habits.
- Personalized budget alerts.
- Visual monthly and category analysis.
- AI-assisted recommendations.
- Supports savings goals.

15. Future Enhancements

- Receipt scanning using OCR.
- Automatic transaction import where supported.
- Voice-based expense entry.
- Predictive monthly expense forecasting.
- Personalized savings challenges.
- Multi-language support, including Indian languages.
- Advanced anomaly and fraud detection.

16. Security & Privacy

Financial data should be treated as sensitive information. The application should use secure authentication, password hashing, authorization checks, encrypted network communication, and appropriate database protections. The system should collect only the data needed for its features and clearly communicate how financial information is stored and used.

17. Conclusion

PocketSmart AI provides an integrated approach to personal budgeting by combining transaction tracking, budget management, analytics, and AI-assisted recommendations. The project can begin as a simple expense-management application and be expanded with machine learning, receipt recognition, forecasting, and conversational assistance. Its primary purpose is to help users understand their own financial behavior and make informed budgeting decisions.

18. Project Summary

Project Name	PocketSmart AI
Project Type	Smart Budget & Recommendation Assistant
Main Purpose	Personal expense and budget management
Key Feature	AI-based spending recommendations
Primary Users	Students, individuals, and families

End of Project Report